

Algorithms 2

Approximating the Maximum Cut Problem

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1. The Max Cut Problem

Problem 1 Max Cut

Given: An undirected graph $G = (V, E)$.

Find: A bipartition $(S, V \setminus S)$ maximising $e_G(S, V \setminus S)$ — the number of edges crossing the cut.

Problem 2 Max Cut

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Theorem 2 For every graph G : $\text{max-cut}(G) \geq \frac{e(G)}{2}$.

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Observation 3 Equivalently: every graph G has a **spanning bipartite subgraph** with $\geq \frac{e(G)}{2}$ edges.

1.2 Proof of the Theorem

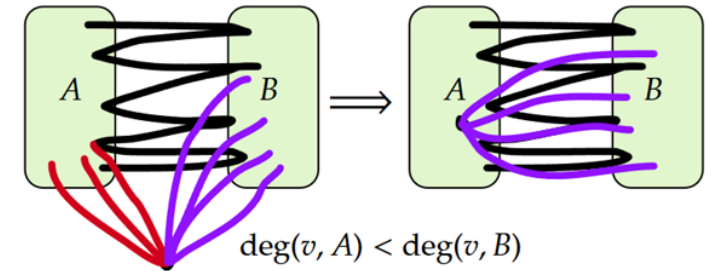
Proof: By induction on $v(G)$. Base cases $v(G) = 1, 2$ are trivial.

Inductive step: Let $v \in V(G)$ be arbitrary.

- Apply the IH to $G - v$:
 - obtain a bipartition $(S, V(G) \setminus (S \cup \{v\}))$ with $\geq \frac{e(G-v)}{2}$ crossing edges.
- Place v on the side where it has the **fewest** neighbours in G
- Then v contributes at least $\deg_G(v)/2$ crossing edges.

Size of resulting cut in G :

$$\geq \frac{e(G-v)}{2} + \deg_G(v)/2 = \frac{e(G)}{2}. \quad \square$$



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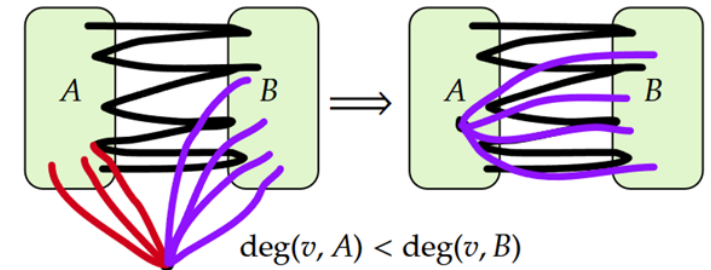
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Theorem 5 The above proof yields a poly-time algorithm forming a $\frac{1}{2}$ -approximation for Max Cut.

2. A Randomised Construction

Algorithm 1: Rand-Cut

```
1:  $A, B \leftarrow \emptyset$ 
2: for each  $v \in V(G)$  do
3:   Toss a fair coin independently
4:   Heads: place  $v$  in  $A$ .   Tails: place  $v$  in  $B$ .
5: end
6: return  $(A, B)$ 
```

Algorithm 2: Rand-Cut

```

1:  $A, B \leftarrow \emptyset$ 
2: for each  $v \in V(G)$  do
3:   Toss a fair coin independently
4:   Heads: place  $v$  in  $A$ .   Tails: place  $v$  in  $B$ .
5: end
6: return  $(A, B)$ 

```

Observation 5 The randomised algorithm achieves an **expected** cut of size $\frac{e(G)}{2}$.

For each edge $e = uv \in E(G)$, define the indicator:

$$X_e := \begin{cases} 1 & e \in E_G(A, B) \\ 0 & e \notin E_G(A, B) \end{cases}$$

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Since the coins are **independent**:

$$\mathbb{E}[X_e] = \mathbb{P}[u \in A \wedge v \in B] + \mathbb{P}[u \in B \wedge v \in A] = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

We may write $e_G(A, B) = \sum_{e \in E(G)} X_e$, so by **linearity of expectation**:

$$\mathbb{E}[e_G(A, B)] = \sum_{e \in E(G)} \mathbb{E}[X_e] = \frac{e(G)}{2}$$

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But how often do we actually get a cut $\geq \frac{e(G)}{2}$?

Theorem 6 The algorithm produces a cut of size $\geq \frac{e(G)}{2}$ with probability $\geq \frac{1}{e(G)+1}$.

(This is bad!)

proof:

- Let $p := \mathbb{P}\left[e_G(A, B) \geq \frac{e(G)}{2}\right]$.
 - **Goal** bound p from below.

Theorem 7 The algorithm produces a cut of size $\geq \frac{e(G)}{2}$ with probability $\geq \frac{1}{e(G)+1}$.

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proof:

- Let $p := \mathbb{P}\left[e_G(A, B) \geq \frac{e(G)}{2}\right]$.
 - **Goal** bound p from below.

Write the expectation by splitting on the cut value:

$$\begin{aligned} \frac{e(G)}{2} = \mathbb{E}[e_G(A, B)] &= \sum_{i \leq \frac{e(G)}{2} - 1} i \cdot \mathbb{P}[e_G(A, B) = i] + \sum_{i \geq \frac{e(G)}{2}} i \cdot \mathbb{P}[e_G(A, B) = i] \\ &\leq \left(\frac{e(G)}{2} - 1\right) \cdot \underbrace{\sum_{i \leq \frac{e(G)}{2} - 1} \mathbb{P}[e_G(A, B) = i]}_{1-p} + e(G) \cdot \underbrace{\sum_{i \geq \frac{e(G)}{2}} \mathbb{P}[e_G(A, B) = i]}_p \end{aligned}$$

Theorem 8 The algorithm produces a cut of size $\geq \frac{e(G)}{2}$ with probability $\geq \frac{1}{e(G)+1}$.

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Rearranging:

$$\frac{e(G)}{2} \leq \left(\frac{e(G)}{2} - 1\right)(1 - p) + e(G) \cdot p$$

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$$\bullet p := \mathbb{P}\left[e_G(A, B) \geq \frac{e(G)}{2}\right] \qquad \bullet \frac{e(G)}{2} \leq \left(\frac{e(G)}{2} - 1\right)(1 - p) + e(G) \cdot p$$

Isolating p :

$$\begin{aligned} \frac{e(G)}{2} &\leq \frac{e(G)}{2} - 1 - p \cdot \left(\frac{e(G)}{2} - 1\right) + e(G) \cdot p \\ 1 &\leq p \cdot \left(e(G) - \frac{e(G)}{2} + 1\right) = p \cdot \left(\frac{e(G)}{2} + 1\right) \\ \Rightarrow p &\geq \frac{1}{\frac{e(G)}{2} + 1} \geq \frac{1}{e(G) + 1}. \quad \square \end{aligned}$$

3. De-randomisation

Goal: Turn the randomised algorithm into a **deterministic** one with the same guarantee.

Key tool: the **method of conditional expectations**.

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Definition 5 Conditional Expectation

For discrete RVs X, Y and value z in the range of Y :

$$\mathbb{E}[X \mid Y = z] := \sum_x x \cdot \mathbb{P}[X = x \mid Y = z]$$

We write $\mathbb{E}[X \mid Y]$ for the **function** $z \mapsto \mathbb{E}[X \mid Y = z]$.

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Theorem 12 Law of Total Expectation

For discrete RVs X, Y :

$$\mathbb{E}[X] = \sum_{z \in \text{range}(Y)} \mathbb{P}[Y = z] \cdot \mathbb{E}[X \mid Y = z]$$

Let $v_1, \dots, v_n$ be an ordering of the vertices. Suppose $v_1, \dots, v_i$ have already been placed in A or B .

By the **Law of Total Expectation**, for vertex v_{i+1} :

$$\mathbb{E}[e_G(A, B) \mid v_1, \dots, v_i \text{ placed}] = \frac{1}{2} \cdot \mathbb{E}[e_G(A, B) \mid \dots, v_{i+1} \in A] + \frac{1}{2} \cdot \mathbb{E}[e_G(A, B) \mid \dots, v_{i+1} \in B]$$

Let:

$$\begin{aligned} \bullet \mathbb{E}_A &= \mathbb{E}[e_G(A, B) \mid \dots, v_{i+1} \in A] & \bullet \mathbb{E}_B &= \mathbb{E}[e_G(A, B) \mid \dots, v_{i+1} \in B] \end{aligned}$$

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Let:

$$\bullet \mathbb{E}_A = \mathbb{E}[e_G(A, B) \mid \dots, v_{i+1} \in A] \qquad \bullet \mathbb{E}_B = \mathbb{E}[e_G(A, B) \mid \dots, v_{i+1} \in B]$$

So:

$$2\mathbb{E}[e_G(A, B) \mid v_1, \dots, v_i \text{ placed}] = \mathbb{E}_A + \mathbb{E}_B$$

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So:

$$2\mathbb{E}[e_G(A, B) \mid v_1, \dots, v_i \text{ placed}] = \mathbb{E}_A + \mathbb{E}_B$$

Observation 8 Either $\mathbb{E}_A \geq \mathbb{E}[e_G(A, B) \mid v_1, \dots, v_i \text{ placed}]$ or $\mathbb{E}_B \geq \mathbb{E}[e_G(A, B) \mid v_1, \dots, v_i \text{ placed}]$.

4. The Deterministic Algorithm

Assuming: $\mathbb{E}_A$ and $\mathbb{E}_B$ can be computed efficiently we have the following algorithm.

Algorithm 3: Derand-Max-Cut

```

1:  $v_1, \dots, v_n \leftarrow$  an arbitrary ordering of  $V$ 
2:  $A, B \leftarrow \emptyset$ 
3: for  $i = 1$  to  $n$  do
4:    $E_A \leftarrow$  Compute  $\mathbb{E}[e_G(A, B) \mid v_1, \dots, v_{i-1} \text{ placed} \wedge v_i \in A]$ 
5:    $E_B \leftarrow$  Compute  $\mathbb{E}[e_G(A, B) \mid v_1, \dots, v_{i-1} \text{ placed} \wedge v_i \in B]$ 
6:   If  $E_A \geq E_B$  place  $v_i$  in  $A$ , else place  $v_i$  in  $B$ 
7: end
8: return  $(A, B)$ 

```

Theorem 13 Derand-Max-Cut is a deterministic poly-time $\frac{1}{2}$ -approximation for Max Cut.

Proof (By induction):

$$\text{I.H. } \forall i < k, \mathbb{E}[e_G(A, B) \mid v_1, \dots, v_i \text{ placed}] \geq \frac{e(G)}{2}$$

Assume without loss of generality that $\mathbb{E}_A \geq \mathbb{E}_B$.

- By assumption we have $\mathbb{E}_A \geq \mathbb{E}[e_G(A, B) \mid v_1, \dots, v_{k-1} \text{ placed}] \geq \frac{e(G)}{2}$.

$\implies$ for $k = n$ we have $e_G(A, B) = \mathbb{E}[e_G(A, B) \mid v_1, \dots, v_n \text{ placed}] \geq \frac{e(G)}{2}$ as required.

For each step i , we need to compute efficiently:

$$\mathbb{E}_A := \mathbb{E}[e_G(A, B) \mid v_1, \dots, v_i \text{ placed} \wedge v_{i+1} \in A]$$

$$\mathbb{E}_B := \mathbb{E}[e_G(A, B) \mid v_1, \dots, v_i \text{ placed} \wedge v_{i+1} \in B]$$

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Let $\tilde{A}, \tilde{B}$ be the **current actual sets** with $v_1, \dots, v_i$ placed. Define:

$$\ell := e(G) - \deg_G(v_{i+1}, \tilde{A}) - \deg_G(v_{i+1}, \tilde{B}) - e_G(\tilde{A}) - e_G(\tilde{B}) - e_G(\tilde{A}, \tilde{B})$$

$\ell = \#$ edges among the **remaining** vertices $v_{i+2}, \dots, v_n$ (still random).

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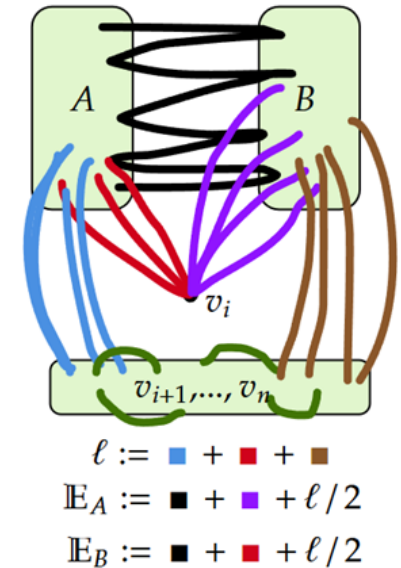
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$\ell = \#$ edges among the **remaining** vertices $v_{i+2}, \dots, v_n$ (still random).

Then:

$$\mathbb{E}_A = e_G(\tilde{A}, \tilde{B}) + \deg_G(v_{i+1}, \tilde{B}) + \frac{\ell}{2}$$

$$\mathbb{E}_B = e_G(\tilde{A}, \tilde{B}) + \deg_G(v_{i+1}, \tilde{A}) + \frac{\ell}{2}$$



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Let $\tilde{A}, \tilde{B}$ be the **current actual sets** with $v_1, \dots, v_i$ placed. Define:

$$\ell := e(G) - \deg_G(v_{i+1}, \tilde{A}) - \deg_G(v_{i+1}, \tilde{B}) - e_G(\tilde{A}) - e_G(\tilde{B}) - e_G(\tilde{A}, \tilde{B})$$

$\ell = \#$ edges among the **remaining** vertices $v_{i+2}, \dots, v_n$ (still random).

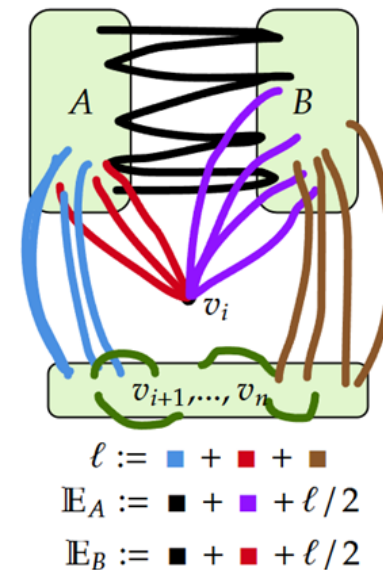
Then:

$$\mathbb{E}_A = e_G(\tilde{A}, \tilde{B}) + \deg_G(v_{i+1}, \tilde{B}) + \frac{\ell}{2}$$

$$\mathbb{E}_B = e_G(\tilde{A}, \tilde{B}) + \deg_G(v_{i+1}, \tilde{A}) + \frac{\ell}{2}$$

Therefore:

$$|\mathbb{E}_A - \mathbb{E}_B| = |\deg_G(v_{i+1}, \tilde{B}) - \deg_G(v_{i+1}, \tilde{A})|$$



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Therefore:

$$\mathbb{E}_A \geq \mathbb{E}_B \iff \deg_G(v_{i+1}, \tilde{B}) \geq \deg_G(v_{i+1}, \tilde{A})$$

$$|\mathbb{E}_A - \mathbb{E}_B| = |\deg_G(v_{i+1}, \tilde{B}) - \deg_G(v_{i+1}, \tilde{A})|$$

Therefore:

$$\mathbb{E}_A \geq \mathbb{E}_B \iff \deg_G(v_{i+1}, \tilde{B}) \geq \deg_G(v_{i+1}, \tilde{A})$$

So we may write:

Algorithm 5: Derand-Max-Cut

```
1:  $v_1, \dots, v_n \leftarrow$  an arbitrary ordering of  $V$ 
2:  $A, B \leftarrow \emptyset$ 
3: for  $i = 1$  to  $n$  do
4:   if  $\deg_G(v_{i+1}, B) \geq \deg_G(v_{i+1}, A)$  place  $v_i$  in  $A$ , else place  $v_i$  in  $B$ 
5: end
6: return  $(A, B)$ 
```

This is the first “algorithm” we saw in those slides.